

Lab 4 Assignment v3

ECE 525.642 FPGA Design using VHDL

VGA Checkerboard Display

For this lab, you will be working through the design of a VGA controller and implement a checkerboard display with a red square that moves along the board based on push-button presses. This will be a 2-week lab and provides a suggested path to completion for weeks 1 and 2. You will submit all of your work at the end of this 2-week lab.

Your design will use the 100MHz board clock to create a VGA controller that drives the VGA Hsync, Vsync, and the four 4-bit color outputs. Your design will use the VGA controller to generate the static checkerboard pattern along with the movable red square. For more detailed information on how the VGA interface should operate, take a look in the Nexy's 4 DDR reference manual and read the Lab 4 lecture notes. For our lab, make sure we only utilize one board clock for all sequential logic and have no latches.

The design should have a checkerboard pattern as shown below in Figure 1, which will have alternating blue/green squares and one red square. The checkerboard pattern should be 15 rows by 20 columns occupying a total visible area of 480x640 pixels. Therefore, each square should be 32x32 pixels. The red square will move around the screen by pressing the up, down, left, and right push buttons. The red square should wrap around the display and move appropriately at the edges. Do not worry about moving the red square diagonally by trying to press two buttons simultaneously.

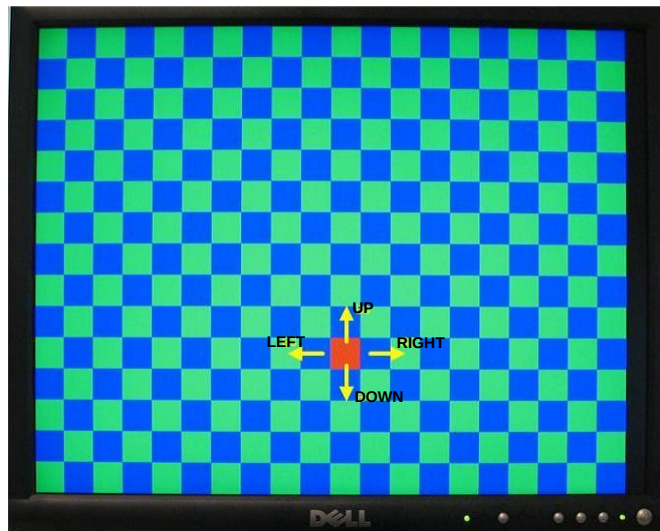


Figure 1 - Checkerboard Pattern Example

Switch 0 is used as an active high reset to the design. Resetting the design should move the red square back to its starting location, which can be any specific choice of your choosing.

The design should use the 7-segment display controller from Lab 3. The display should show the X coordinates on the displays 3 and 2 and the Y coordinates on displays 1 and 0. The x-coordinates will be from 0x00 to 0x13 and the y-coordinates from 0x00 to 0x0E. For example, (x,y) of (0x00, 0x00) would be displayed as '0000' and (0x11, 0x0C) would be shown as '110C.' The origin should be (0x00, 0x00) and be in the upper left.

Summary of Inputs, Outputs, and Signals:

Inputs:

- Switch 0: Active high reset
- Button Up (BTNU): Move the red square up one position per button press
- Button Down (BTND): Move the red square down one position per button press
- Button Left (BTNL): Move the red square left one position per button press
- Button Right (BTNR): Move the red square right one position per button press
- Clock: System Clock, 100MHz

Outputs:

- VGA Ports: To drive VGA display
 - VGA_R, VGA_G, VGA_B: Red, Green, Blue color signals, each 4-bits wide
 - VGA_HS, VGA_VS: Horizontal and Vertical Syncs, each 1-bit wide
- 7-segment display: To display x-y coordinates (SEG7_CATH, AN)
 - SEG7_CATH: Cathode signals, 8-bits wide
 - AN: Anode signals, 8-bits wide

Notes:

- Read the Nexy's reference manual section on VGA display before beginning this lab then take a look at the Lab 4 hints document for a starting path.
- Make sure to debounce **ALL** of the pushbuttons! Test the debouncing by quickly pressing and releasing the buttons. The red square should only move **ONCE** per button press! The movement can happen on either the press of the button or release but should only happen for ONE of them!
- You should generate a 25MHz enable signal derived from the 100MHz system clock. ALL sequential processes should use the 100MHz clock.
- You may need to 'reset' the monitor the first time you connect your Nexy's development board to the monitor. This allows the monitor to synchronize to the Hsync/Vsync signals. This is sometimes found in the monitor's menu or setup menu.



Task 1 (Work on by 1st week)

Start the first week by building up a block diagram of the VGA controller prior to starting the major coding effort. Try to think about estimates for resources needed (consider flip-flops, clocks, I/O).

Make sure to break down the VGA controller block diagram into subcomponents with detail similar to the previous block diagrams seen in this course. The block diagram should consist of counters, timers, comparators, latches, muxes, and ports. The diagram should be detailed enough to gain an understanding for how you would implement this design in the FPGA.

After working out the block diagram, implement just the VGA controller to display a static checkerboard pattern. Do not worry about the red square and associated logic to move the square. You can follow the Lab 4 hints to accomplish this with a solid red background.

Task 2

After getting a basic VGA controller working, the next task is to add in the logic to draw the checkerboard pattern. Think about how the powers of two can be used to generate this pattern. A 32x32 pattern will alternative the 5th bit of the vertical and horizontal counters. You can use this to generate a simple conditional to determine if the current output color should be blue or green.

Once the checkerboard is designed, work on adding in the red square which has the position indicated by the 7-seg controller and internal registers for the x/y position of the red square block. You can use the red square position and the upper bits of the horizontal and vertical counters to determine if the block should be red. Otherwise, use the logic above to draw the green/blue color. To help visualize this, try drawing out a few squares of the display and indicate the binary values at the edges of each square. This can help show how this logic should function.

The last step is to then add in the debounce logic along with the logic to increment/decrement the x/y red square position. The lectures in the course provide some background on this and are worth reviewing prior to starting this work. To test this, verify that pressing a button once does not cause the square to move more than once.

Lastly, test your edge cases for the design (wrap around and reset cases) and document the design along with comparison to initial design estimates with actuals. Put everything together that is discussed in the section below and submit the lab.

To Submit

Files to Submit (minimum), zipped together as Lab4_<last_name>.zip:

- VHD, XDC, BIT files for Lab 4
- Complete Synthesis Results
- Report summarizing the resource utilization for Lab 4, overall design principles, and completed block diagram incorporating any changes from Task 1

Note: The Grading sheet provide on the next page provides a good overview of the key points of the design and what the instructor will look for when grading. Please note that this is a guide and isn't comprehensive. Please read over the lab assignment carefully to verify all the proper documents are submitted.



Grading Sheet

Lab 4/Module 5: VGA Controller and Checkboard Pattern

Lab 4

Student:

Grade	Out of	Notes
	100	
	10	VGA Checkerboard display is correct (green/blue checkered pattern) without missing or miscolored pixels
	5	Screen is clear and not flickering (correct Hsync/Vsync)
	10	Switch0 resets the design and brings red square back to starting point
	10	X-Y coordinates displayed correctly on 7-segment display
	5	Red Square moves in correct direction by button presses
	10	Red Square only moves one square location per button press
	10	Red square correctly wraps at screen edges
	15	Block diagram of design
	15	Quality of VHDL
	10	Summary and detailed analysis including Initial Design Document